INCH-POUND

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COMMERCIAL ITEM DESCRIPTION

LATHE, BENCH AND JEWELER'S, HORIZONTAL SPINDLE

The General Services Administration has authorized the use of this commercial item description for all federal agencies.

1. SCOPE. This commercial item description (CID) covers horizontal spindle, bench, and jeweler's lathes used in operations that require small metal workpiece turning, drilling, facing, or precise secondary finishing.

2. CLASSIFICATION. The lathes shall be of the following types and sizes.

2.1 Type. The type to be furnished shall be as specified (see 7.3(b)).

Type I - bench lathe
Type II - jeweler's lathe

2.2 Size. The size to be furnished shall be as specified (see 7.3(c)).

Size - see tables I and II

Beneficial comments, recommendations, additions, deletions, clarifications, etc. and any data that may improve this document should be sent to: STDZNMGT@dla.mil or Defense Supply Center Richmond (DSCR), ATTN: DSCR-VEB, 8000 Jefferson Davis Highway, Richmond, VA 23297-5616.

AMSC N/A FSC 3416

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TABLE I. Type I (bench) machine sizes.

Size	Swing over bed (inches)	Center-to-center distance (inches)
0308	3	8
0317	3	17
0511	5	11
0710	7	10
0715	7	15
0819	8	19
0833	8	33
1118	11	18
1120	11	20
1125	11	25
1224	12	24
1236	12	36
1334	13	34
1336	13	36
1340	13	40

Note: The sizes shown are minimum requirements.

TABLE II. Type II (jeweler's) machine sizes.

Size	Swing over bed (inches)	Center-to-center distance (inches)
0304	3	4
0607	6	7
0608	6	8
0611	6	11
0612	6	12
0807	8	7
0813	8	13
0815	8	15
0817	8	17

Note: The sizes shown are minimum requirements.

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3. SALIENT CHARACTERISTICS

3.1 General requirements. The machine shall be new and one of the manufacturer's current models capable of operation in accordance with the requirements herein. All parts subject to wear, breakage, or distortion shall be accessible for adjustment, replacement, and repair.

3.2 Components. The lathes shall include, as a minimum, the following components.

3.2.1 Bed. The bed shall support all the components of the lathe and shall have provisions for mounting to a benchtop or cabinet.

3.2.2 Ways. The ways shall be hardened and precision ground.

3.2.3 Spindle. The spindle shall be capable of multiple speeds through belt or gear drives, as specified (see 7.3(d)). The spindle shall have sealed and permanently lubricated bearings. If required, the spindle shall be equipped with the following (see 7.3(e)):

a. Foot-operated, hand-operated, or automatic brake.
b. Reverse rotation capability.
c. Powered collet closer.

3.2.4 Material holder. The lathe shall have material holding mechanisms such as draw-in collets, multiple-jaw chucks, or shank-holding Morse tapers, as specified (see 7.3(f)). The spindle shall accommodate any of the three material holders.

3.2.5 Tailstock. The lathe shall have a tailstock that is adjustable and lockable along the ways. The tailstock shall accommodate material holding centers and Morse taper tool inserts for drilling and boring operations.

3.2.6 Type I (bench) lathes.

3.2.6.1 Tool holder. The tool holder on the type I lathe shall be mounted on a carriage and cross-slide. The carriage shall hold the cross-slide and the tool post. The carriage shall move along the workpiece axis on the ways by hand crank or power feed, as specified (see 7.3(g)). Unless otherwise specified (see 7.3(h)), the power-feed lead screw shall be capable of being coupled to the spindle to coordinate feed and rotation for cutting threads. The cross-slide shall provide tool movement along the axis perpendicular to the workpiece axis and through the swing of the lathe. The cross-slide shall be equipped with a compound rest that shall allow the 360-degree adjustment of the tool with respect to the workpiece axis. When specified (see 7.3(i)), the lathe shall be provided with a multiple tool-holding turret that can be installed on the ways in place of the tailstock. The turret shall possess a minimum of four tool positions, and shall be indexable with independently adjustable stops for each turret position. The turret shall have a means to feed and retract the indexed tool into the workpiece.

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3.2.6.2 Controls. Type I lathe controls shall, as a minimum, consist of spindle speed, tool feed, cutting depth, and start/stop. Unless otherwise specified (see 7.3(j)), control adjustments shall be accomplished by hand wheels, locking levers, or control knobs. Type I lathes with a powered tool feed lead screw shall have a means to adjust the tool feed rate and thread cutting pitch.

3.2.7 Type II (jeweler's) lathes.

3.2.7.1 Tool holder. As a minimum, type II lathes shall be equipped with slides that allow tool movement along the workpiece axis and perpendicular to the workpiece axis to allow turning, facing, and boring. When specified (see 7.3(k)), the slides shall be powered. When specified (see 7.3(l)), the lathe shall be provided with a multiple tool-holding turret that can be installed on the ways. The turret shall possess a minimum of four tool positions and shall be indexable with independently adjustable stops for each turret position. The turret shall have a means to feed and retract the indexed tool into the workpiece.

3.2.7.2 Controls. The type II lathe controls shall, as a minimum, consist of spindle speed, tool feed, cutting depth, and start/stop. Unless otherwise specified (see 7.3(m)), control adjustments shall be accomplished by hand wheels, locking levers, or control knobs. When specified (see 7.3(n)), controls shall be provided that allow tool depth and feed times to be preset and then controlled automatically upon manual start. When specified (see 7.3(o)), the jeweler's lathe shall provide automatic stop and workpiece ejection at the end of the turning operation.

3.2.8 Lubrication system. Unless otherwise specified (see 7.3(p)), machine components requiring continuous lubrication shall be permanently lubricated or lubricated by automatic means. Machine components requiring periodic manual lubrication shall be readily accessible.

3.2.9 Coolant system. When specified (see 7.3(q)), the bench lathe shall be equipped with a tool coolant system. The coolant system shall consist of a reservoir or sump, powered pump, chip tray, and nozzle(s).

3.2.10 Electrical system. The electrical system shall conform to the American National Standards Institute/National Fire Protection Association (ANSI/NFPA) 79, "Electrical Standard For Industrial Machinery", or International Electrotechnical Commission (IEC) 204.1, "Safety of Machinery - Electrical Equipment of Machines Part 1: General Requirements For". Unless otherwise specified (see 7.3(r)), the bench lathe shall operate from a dual 230/460-volt, single-phase, 60-hertz circuit.

3.2.11 Electrical motors. The spindle drive motor shall have sealed and permanently lubricated ball or roller bearings, shall be energy efficient, and shall conform to National Electrical Manufacturer's Association (NEMA) MG-1, "Motors and Generators". Unless otherwise specified (see 7.3(s)), the motors shall have drip-proof enclosures.

3.2.12 Optional components. Optional components shall be provided as specified (see 7.3(t)).

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3.3 Performance. Unless otherwise specified (see 7.3(u)), the lathes shall meet the performance requirements of table III when tested in accordance with sections 5.3 through 5.3.4 herein.

TABLE III. Performance requirements.

Parameter	Type I	Type II
Collet run-out, inch	0.0002	0.0002
Spindle run-out, inch	0.0005	0.0005
Cylinder turn: Roundness, inch Taper, inch	 0.001 0.002	
Facing flatness, inch		0.001

3.4 Machine dimensions and weight. Unless otherwise specified (see 7.3(v)), the maximum dimensions (length, width, and height) and machine weight shall not exceed the restrictions specified.

3.5 Safety and health requirements. The machine shall be designed and manufactured in accordance with ANSI B11.6, "Safety Requirements For Manual Turning Machines With or Without Automatic Control". The manufacturer shall ensure that the machine and all equipment and accessories used on the machine shall be in compliance with 29 CFR 1910, "Occupational Safety and Health Standards". If a conflict arises between the ANSI and CFR standards, the CFR standard shall apply.

3.6 System of units. The U.S. Customary System of Units (US) or the International System of Units (SI) shall be used to graduate measuring and indicating devices. When only one system of graduation is acceptable, the particular graduation required shall be as specified (see 7.3(w)). Regardless of the measurement system used, all measuring and indicating devices on the machine shall be graduated in the same system. When specified (see 7.3(x)), measuring and indicating devices shall be graduated in both the US and SI system of measurements.

3.7 Nameplate. A nameplate shall be securely attached to the machine. Unless otherwise specified (see 7.3(y)), the nameplate shall contain the following information:

a. Nomenclature.
b. Manufacturer's name.
c. Manufacturer's model designation.
d. Machine type and size classification (see 2.1).
e. Power input (volts, total amperes, phase, frequency).
f. Short-circuit/over-circuit rating.
g. Contract or order number.
h. National stock number.
i. Date of manufacture.

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3.8 Lubricant plate or chart. When specified (see 7.3(z)), a lubricant plate or chart shall be attached to the machine. The information provided on the plate shall include:

a. Points of lubricant application.
b. Servicing interval.
c. Type of lubricant(s) with SAE number or lubricant identifier.

4. REGULATORY REQUIREMENTS

4.1 Recovered materials. The offeror/contractor is encouraged to use recovered materials to the maximum extent practicable, in accordance with paragraph 23.403 of the Federal Acquisition Regulation (FAR).

5. PRODUCT CONFORMANCE PROVISIONS

5.1 Product conformance. The products provided shall meet the salient characteristics of this CID, conform to the producer's own drawings, specifications, standards, and quality assurance practices, and be the same product offered for sale in the commercial marketplace. The government reserves the right to require proof of such conformance.

5.2 Inspection. The bench lathe shall be inspected to determine compliance with all requirements specified in this CID.

5.3 Testing. Unless otherwise specified (see 7.3(aa)), the bench lathe shall be tested in accordance with sections 5.3.1 through 5.3.4.

5.3.1 Test setup and conditions. Unless otherwise specified (see 7.3(bb)), the manufacturer shall be responsible for supplying all tooling and materials and shall conduct the tests required by the government. The manufacturer shall provide test instruments and equipment with calibration traceable to the National Institute of Standards and Technology (NIST) and shall provide certification of calibration as specified (see 7.3(cc)). Unless otherwise specified (see 7.3(dd)), the machine shall be tested under environmental conditions deemed appropriate by the manufacturer.

5.3.2 Operational test. The lathe shall be operated in accordance with the manufacturer's standard operating procedures for a period of one hour. Proper operation of all controls and features shall be verified during the test period.

5.3.3 Performance tests. Unless otherwise specified (see 7.3(ee)), the performance test shall consist of spindle run-out, cylindrical turning, and facing tests.

5.3.3.1 Spindle run-out. The spindle run-out test shall consist of obtaining the measurement data for the parameters for spindle and collet run-out as specified in 5.3.3.1.1 and 5.3.3.1.2. The measurements shall not exceed the performance requirements shown in table III for the particular test. The measuring instrument(s) shall be the manufacturer's choice.

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5.3.3.1.1 Spindle center runout. The total indicator reading for a tool room lathe shall be 0 to 0.0004. The tolerances for the engine room lathes total indicator reading shall be as follows:

a. 12 to 18 inches inclusive shall be 0 to 0.0005.
b. 20 to 32 inches inclusive shall be 0 to 0.00075.
c. 40 to 72 inches inclusive shall be 0 to 0.001.

5.3.3.1.2 Collet chuck runout. The tolerances 1 inch from the collet chuck shall be 0 to 0.001. No collet chucks are used on the 40 to 72 inch engine lathes.

5.3.3.2 Cylindrical turning test. Type I lathes shall undergo the cylindrical turning as specified in 5.3.3.2.1 and 5.3.3.2.2. The test workpiece shall be AISI 1020 steel. The workpiece length and diameter shall be 1/2 the maximum swing and center-to-center capacity of the lathe. The tool choice and depth of test cut shall be at the manufacturer's discretion. The length of the test cut shall be 3/4 of the length of the test workpiece. The roundness and taper over the test cut length measurements shall meet the requirements shown in table III.

5.3.3.2.1 Lathe turn, round. The lathe shall turn around the workpiece with the work mounted in the chuck. The tolerances for a tool room lathe shall be 0.0003. The tolerances for the engine lathes shall be as follows:

a. 12 to 18 inches inclusive shall be 0.0004.
b. 20 to 32 inches inclusive shall be 0.0008.
c. 40 to 72 inches inclusive shall be 0.001.

5.3.3.2.2 Lathe turn, cylindrical. The lathe shall turn cylindrically with a 12-inch piece of work mounted on centers. The tolerances for a tool room lathe shall be 0.0004. The tolerances for the engine room lathes shall be as follows:

a. 12 to 18 inches inclusive shall be 0.0008.
b. 20 to 32 inches inclusive shall be 0.001.
c. 40 to 72 inches inclusive shall be 0.001.

5.3.3.3 Facing test. Type II lathes shall undergo a facing test. The test workpiece shall be AISI 1020 steel. The workpiece length and diameter shall be half the maximum capacity of the lathe. The tool choice and depth of cut shall be at the manufacturer's discretion. Facing flatness shall not exceed the value shown in table III.

5.3.4 Optional tests. Optional tests shall be conducted as specified (see 7.3(ff)).

5.4 Acceptance. Unless otherwise specified (see 7.3(gg)), preliminary and final acceptance tests shall be conducted at the manufacturer's site. The results of the performance tests shall be compared to the performance requirements shown in table III. Failure of the system to meet the performance requirements shall be cause for rejection.

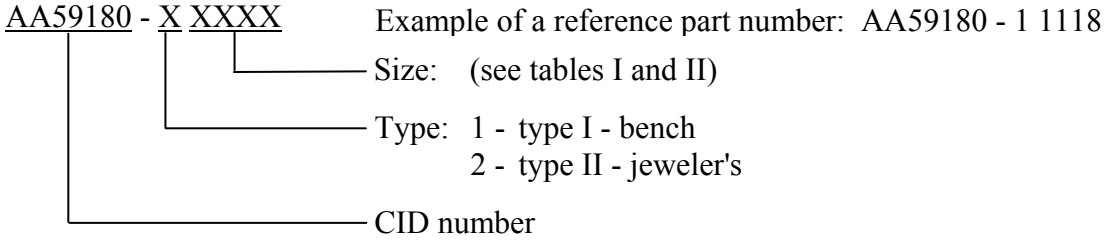
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6. PACKAGING

6.1 Preservation, packing, and marking. Preservation, packing, and marking shall be as specified in the acquisition order (see 7.3(hh)).

7. NOTES

7.1 Part or identification number (PIN). The following PIN procedure is for government purposes and does not constitute a requirement for the contractor.



7.2 Sources of documents.

7.2.1 CFR and FAR. Copies of CFR and FAR may be obtained from the Superintendent of Documents, P.O. Box 371954, Pittsburgh, PA 15250-7954. Electronic copies of CFR documents may be obtained from http://www.access.gpo.gov/. Electronic copies of FAR documents may be obtained from http://www.arnet.gov/far/.

7.2.2 NIST handbooks. Copies of NIST handbooks may be obtained from the Office of Weights and Measures, 100 Bureau Drive, Stop 2600, Gaithersburg, MD 20899-2600. Electronic copies of NIST handbooks may be obtained from http://ts.nist.gov/.

7.2.3 ANSI standards. Copies of ANSI standards may be obtained from the American National Standards Institute, 25 West 43rd Street, 4th Floor, New York, NY 10036. Electronic copies of ANSI standards may be obtained from http://web.ansi.org/.

7.2.4 NEMA standards. Copies of NEMA standards may be obtained from the National Electrical Manufacturers Association, 1300 North 17th Street, Suite 1847, Rosslyn, VA 22209. Electronic copies of NEMA standards may be obtained from http://www.nema.org/.

7.2.5 NFPA standards. Copies of NFPA standards may be obtained from the National Fire Protection Association, 1 Batterymarch Park, P.O. Box 9101, Quincy, MA 02269-9101. Electronic copies of NFPA standards may be obtained from http://www.nfpa.org/.

7.2.6 IEC standards. Copies of IEC standards may be obtained from IEC Central Office, 3, rue de Varembé, P.O. Box 131, CH-1211 Geneva 20, Switzerland. Electronic copies of IEC standards may be obtained from http://www.iec.ch/.

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7.3 Ordering data. The acquisition order should specify the following:

a. CID document number, revision, and CID PIN.
b. Type of bench lathe required (see 2.1).
c. Size of lathe required (see 2.2).
d. Spindle speeds and belt or gear drive, as specified (see 3.2.3).
e. Foot operated, hand operated, or automatic brake; reverse rotation; powered collet closer, if required (see 3.2.3).
f. Draw-in collet, multiple-jaw chuck, or taper, as specified (see 3.2.4).
g. Hand crank or power feed for type I lathes, as specified (see 3.2.6.1).
h. Lead screw for type I lathes, if different (see 3.2.6.1).
i. Tool-holding turret for type I lathes, if required (see 3.2.6.1).
j. Type I control adjustments, if different (3.2.6.2).
k. Powered slides for type II lathes, if required (see 3.2.7.1).
l. Tool-holding turret for type II lathes, if required (see 3.2.7.1).
m. Type II control adjustments, if different (see 3.2.7.2).
n. Automatic tool depth and feed time controls for type II lathes, if required (see 3.2.7.2).
o. Automatic stop and workpiece ejection for type II lathes, if required (see 3.2.7.2).
p. Lubrication system, if different (see 3.2.8).
q. Tool coolant system, if required (see 3.2.9).
r. Electrical system, if different (see 3.2.10).
s. Motor enclosure, if different (see 3.2.11).
t. Optional components, as specified (see 3.2.12).
u. Performance requirements, if different (see 3.3).
v. Machine dimensions and weight, if different (see 3.4).
w. System of units required, when only one system is acceptable (see 3.6).
x. Dual system of units, if required (see 3.6).
y. Nameplate information, if different (see 3.7).
z. Lubrication plate or chart, if required (see 3.8).
aa. Testing, if different (see 5.3).
bb. Test responsibility, if different (see 5.3.1).
cc. Calibration certification, as specified (see 5.3.1).
dd. Environmental conditions, if different (see 5.3.1).
ee. Performance tests, if different (see 5.3.3).
ff. Optional tests, as specified (see 5.3.4).
gg. Acceptance, if different (see 5.4).
hh. Preservation, packing, and marking requirements (see 6.1).

7.4 Subject term (key word) listing.

drilling
facing
finishing
turning

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MILITARY INTERESTS:	CIVIL AGENCY COORDINATING ACTIVITY:
Custodians: Army - AR Navy - SH Air Force - 99	GSA - FSS
Review Activities: Army - AV Navy - MC Air Force - 84	Preparing Activity: DLA - GS6 (Project 3416-0280)

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